

Q109 Note :- Apart from print statement when we do other calculations we need Operators or whatever you do you need operators.

Ex:- Print ("a")

Without Operators  
We can't write a  
code except print  
statement

this print statement doesn't required Operators  
but other than that any Operation or  
Calculation you do you need Operators.

## 5th topic Operators

Operators :- It is a special symbol perform certain Operation  
between Operands.

Ex:-  $a = 3$

= Operator     $a, 3$  Operands

$z = x + y$

= + Operator     $z, x, y$  Operands

## Types of Operators

- 1) Arithmetic Operators :- used for basic math Operations
  - + - \* / % // \*\*
  - // - floor division → removes decimal & gives integer
  - % - Modulus - remainder
  - \*\* (Exponentiation - Power)
- 2) Comparison / relational Operators
  - > < <= >= == !=
  - Compare values & return True/False



3) Logical Operators : Combine Conditions  
 and or not → (True if both are true)  
 → (True if at least one is true)  
 → (Negates the condition)

4) Assignment Operators : Assign Values to Variables  
 = += -= \*= /= %= \*\*=

5) bitwise Operators  
 & | ~ >> << ^

6) identity Operators  
 is isnot

7) Membership Operators  
 in notin

Assignment Operator  
 = Assign

+ = (Add and assign →  $x += 2$  means  $x = x + 2$ )  
 - = (Subtract and assign)  
 \* = (Multiply and assign)  
 / = (Divide and assign)

is, isnot

8) Bitwise Operators - Work at Binary level

& (AND)

| (OR)

~ (NOT → flips bits)

<< (Left shift)

>> (Right shift)

^ (XOR)

9) Membership Operators - Check membership in Sequence

in (Returns True if value is present)

not in (Returns True if value is not present)

ex: `Print(8 in [1, 2, 3])`

o/p True

~~assign 2 to x~~  
~~add 2 to x~~ and store in x  
 $x = 5$   
 $x += 2$   
~~Print(x)~~  
 o/p: 7

3)  $\frac{8}{2}$  (6 - Quotient if want this use floor div)  
 2 - remainder



\*) identity Operators - Compare object reference  
is (true if both refers to same object)  
is not (true if not same object)

Arith - Math  
Comp - checking values - this Comparison Operator we  
Logical - conditionals Can't able create individual  
we have to use with Conditional statement.

Assignment → shortcut for updating values

Bitwise - binary Operations

Membership → in/not in

identity → object references.